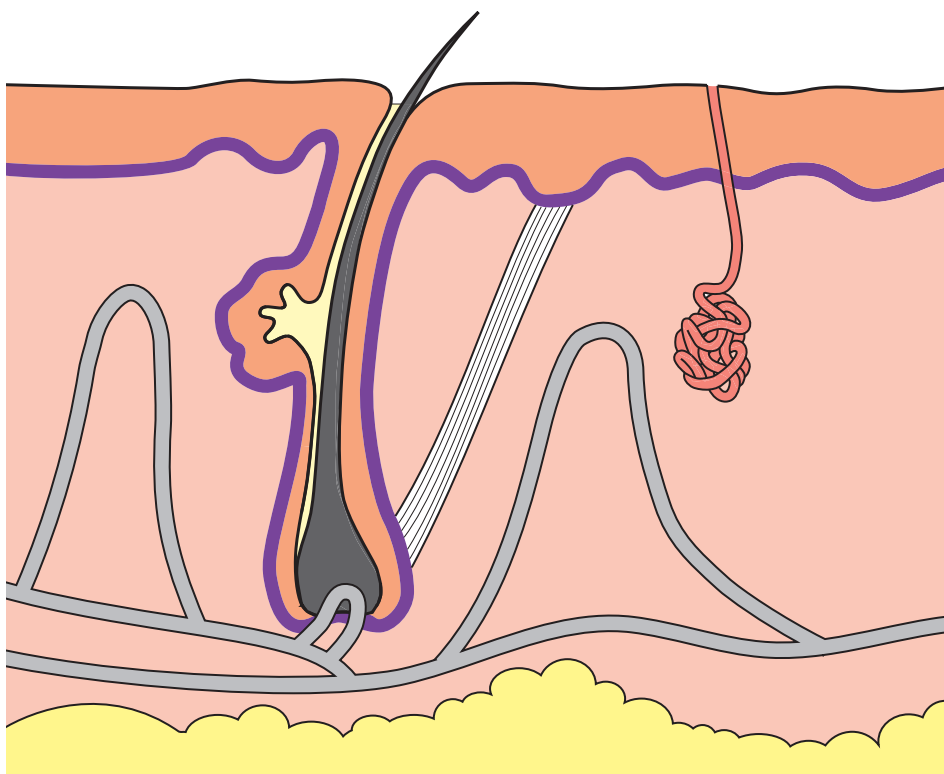


Answer **all** questions.

1. **Image 1.1** shows a section through human skin.

Image 1.1



- (a) On **Image 1.1**, use arrows to label:

[2]

- I. the sweat duct
- II. a blood vessel



- (b) Scientists investigated the effect of air temperature on both the skin temperature and core body temperatures of five volunteers. Core body temperature is the temperature of the internal organs of the body.
- The five volunteers were placed in a temperature controlled laboratory at -20°C .
 - The volunteers wore bathing suits and were kept at the temperature for 5 minutes after which their skin and core temperatures were recorded.
 - The experiment was then repeated at other air temperatures over the next six days, each day at a different air temperature.

The skin temperatures at different air temperatures are shown in **Table 1.2**. The mean core body temperatures are shown on **Graph 1.3**.

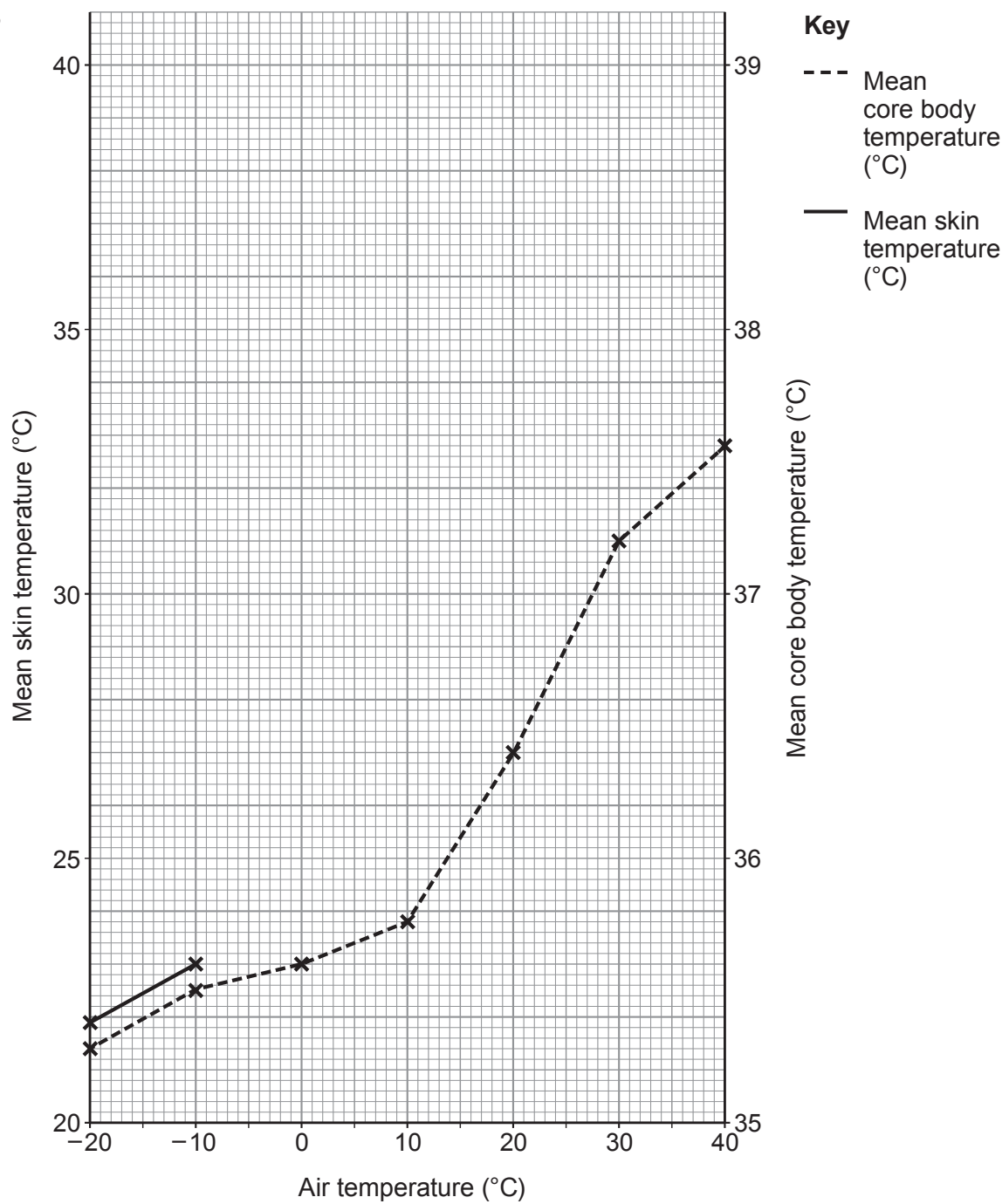
Table 1.2

Air temperature (°C)	Skin temperature of volunteers (°C)					Mean skin temperature (°C)
	Volunteer number					
	1	2	3	4	5	
−20	21.4	22.0	22.6	21.8	21.5	21.9
−10	23.3	23.2	22.8	22.9	23.0	23.0
0	23.8	24.2	24.4	23.5	23.9	24.0
10	25.7	25.3	25.9	25.2	25.2	25.5
20	28.3	28.1	28.5	27.9	27.8	28.1
30	33.0	32.3	32.7	32.4	32.2	32.5
40	38.5	40.2	39.3	40.2	39.1	

- (i) **Complete Table 1.2** by calculating the mean skin temperature of the five volunteers at a temperature of 40°C . [1]
- (ii) I. Plot the **mean skin temperature** against the **left-hand** y-axis on **Graph 1.3**. The first two points have been plotted for you. [2]
- II. **Join the plots with a ruler.** [1]
- The mean core body temperature is already plotted against the right-hand y-axis.



Graph 1.3



III. From **Graph 1.3** determine the mean core body temperature when the air temperature is 25°C. [1]

Mean core body temperature when air temperature is 25 °C = °C

IV. From **Graph 1.3** state the relationship between the skin temperature and the core body temperature. [1]

.....

.....



(c) (i) Place a tick (✓) in **one** of the boxes below to show the processes which occur in the skin when air temperature is **low**. [1]

☐

blood vessels constrict, sweating reduces, hairs lowered on skin surface, shivering occurs

☐

blood vessels dilate, sweating reduces, hairs raised on skin surface, shivering occurs

☐

blood vessels constrict, sweating reduces, hairs raised on skin surface, shivering occurs

☐

blood vessels constrict, sweating increases, hairs raised on skin surface, shivering occurs

(ii) Place a tick (✓) in **one** of the boxes below to show the processes which occur in the skin when air temperature is **high**. [1]

☐

blood vessels dilate, sweating increases, hairs lowered on skin surface, shivering occurs

☐

blood vessels dilate, sweating increases, hairs lowered on skin surface, no shivering

☐

blood vessels constrict, sweating increases, hairs lowered on skin surface, no shivering

☐

blood vessels dilate, sweating decreases, hairs lowered on skin surface, no shivering

☐

10

